SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

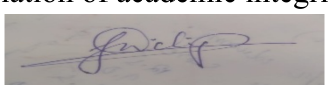
Total Marks: 50

Question Count: 4

Code of Conduct

I Jani Dilip (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

1. Concept Map 1: Cloud Computing Fundamentals

Cloud Computing

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet. It allows users to access resources on demand without owning or maintaining physical infrastructure.

Cloud Computing

Definition

- Delivery of computing resources over the Internet.
- Provides services on a pay-as-you-go basis.
- Eliminates the need for owning physical infrastructure.

Characteristics

- On-demand self-service: Users can access resources whenever needed.
- Broad network access: Services are available through the Internet.
- Resource pooling: Resources are shared among multiple users.
- Rapid elasticity: Resources can be increased or decreased quickly.
- Measured service: Users pay only for the resources they use.

Benefits

- Reduces IT costs.
- Easy scalability and flexibility.
- High availability and reliability.
- Faster deployment of applications.
- Automatic software updates and maintenance.

Limitations

- Security and privacy concerns.
- Dependence on Internet connectivity.
- Possible service downtime.
- Vendor lock-in.
- Limited control over infrastructure.

2. Concept Map 2: Evolution of Cloud Computing

Evolution of Cloud Computing

Cloud computing evolved through advancements in hardware, networking, software technologies, and distributed computing.

Evolution of Cloud Computing

Hardware Evolution

- Mainframe computers.
- Client-server architecture.
- Virtual machines.
- High-performance servers.

Internet Software Evolution

- Development of the World Wide Web.
- Web applications.
- APIs and Web Services (SOAP and REST).
- Browser-based software.

Distributed Systems

- Grid computing.
- Cluster computing.
- Utility computing.
- Parallel processing.

Service Delivery

- Computing resources delivered over the Internet.
- On-demand access.
- Pay-as-you-go pricing.
- Supports public, private, and hybrid cloud models.

Relationship

- Hardware improvements enabled virtualization.
- Internet technologies enabled online service access.
- Distributed systems enabled resource sharing.

3. Concept Map 3: Service Models

Cloud Service Models

Cloud computing offers different service models depending on the level of control required by users.

Concept Map (Theory + Points)

Cloud Service Models

IaaS (Infrastructure as a Service)

- Provides virtual machines, storage, and networking.
- User Responsibilities

PaaS (Platform as a Service)

- Provides a platform for developing and deploying applications.
- Provider Responsibilities
- User Responsibilities

SaaS (Software as a Service)

- Provides ready-to-use software over the Internet.
- Provider Responsibilities
- User Responsibilities

XaaS (Everything as a Service)

- Delivers various IT services through the cloud.

Relationship

- IaaS provides infrastructure.
- PaaS provides a development platform.
- SaaS provides complete software.
- XaaS extends cloud services to almost every IT resource.

4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud

Virtualization and Web Services

Virtualization and Web Services are two key technologies that make cloud computing possible by improving resource utilization, communication, and scalability.

Virtualization

- Creates multiple virtual machines on one physical server.
- Improves hardware utilization.
- Enables resource pooling.
- Supports server consolidation.
- Provides isolation between applications.
- Enables quick provisioning of resources.

Web Services

- Allow communication between different applications.
- Use Internet protocols such as HTTP.
- Support APIs.
- Common types:
- Enable interoperability between different platforms.

Support for Cloud Computing

- Virtualization provides
- Web Services provide

Outcome

- Together, virtualization and web services enable cloud deployment by providing: